MATH 205

Linear Algebra

LECTURE 5 · PROPERTIES OF DETERMINANTS

How det(AB), det(A^T), row and column operations, and row swaps all behave — the algebraic toolkit that makes determinants usable without brute-force expansion.

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WELCOME BACK

Where Today Fits

Last week

- Reduced row-echelon form, Gauss-Jordan elimination, and free variables
- The geometry of Ax = b, plus the 2×2 inverse and its determinant, ad − bc
- First look at the master theorem

Today

- How determinants behave under products, transposes, and inverses
- What row and column operations do to a determinant — and what they don't
- Why swapping two rows or columns always flips the sign
- Why det(A) + det(B) ≠ det(A+B) in general, and the one case where it does add

Course Roadmap

1. Systems of Linear Equations
2. Matrices & Matrix Algebra
3. **Determinants**
4. Vector Spaces
5. Eigenvalues & Eigenvectors

These properties are the whole point — once you know them, you can evaluate and manipulate determinants without ever brute-forcing a big expansion.

3.1 PROPERTIES OF DETERMINANTS

Determinants Have Structure

You already know det(A) = ad − bc for a 2×2 matrix. What matters far more is how det behaves under products, transposes, and row operations — that structure is what makes determinants usable at any size.

1. Products & the Identity

1 If A and B are square matrices of the same size, then det(AB) = det(A) det(B).

Check it for 2×2 matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \det(A) = a_{11}a_{22} - a_{12}a_{21}$$

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad \det(B) = b_{11}b_{22} - b_{12}b_{21}$$

Multiplying the two expressions out confirms det(A)det(B) exactly matches the determinant of the product AB — same four terms, just grouped differently.

2 det(I) = 1, where I is the identity matrix — true for the identity of any order.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \det(I) = 1$$

Anton & Rorres, Elementary Linear Algebra — p.97 (8th Ed.) / p.95 (7th Ed.)

2. Invertibility: Revisited

Last lecture: if A is invertible, det(A) ≠ 0. The product rule pins down exactly what det(A^{-1}) equals.

$$A^{-1}A = I \implies \det(A^{-1}A) = \det(I) = 1 \implies \det(A^{-1})\det(A) = 1$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

det(A) can never be zero when A is invertible — if it were, this formula would divide by zero, which is exactly why det(A) ≠ 0 is required.

3. Determinants Don't Care About Transposing

$$\det(A) = \det(A^T)$$

For a 2×2 matrix, this is immediate — transposing just swaps the off-diagonal entries, which the determinant formula treats symmetrically:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

$$\det(A) = \det(A^T) = a_{11}a_{22} - a_{21}a_{12}$$

Anton & Rorres, Elementary Linear Algebra — p.89 (8th Ed.) / p.86 (7th Ed.)

4. Repeated Rows or Columns Collapse to Zero

If a square matrix A has two identical rows or two identical columns, then det(A) = 0.

Identical columns

$$\begin{vmatrix} a & a \\ b & b \end{vmatrix} = ab - ba = 0$$

Identical rows

$$\begin{vmatrix} a & b \\ a & b \end{vmatrix} = ab - ba = 0$$

This makes intuitive sense too: two identical rows mean the rows are linearly dependent — the “system” collapses, and a matrix built from it can't be invertible.

5. Adding Multiples of Rows or Columns Together Changes Nothing

Starting from the definition: $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

Add column 2 into column 1:

$$C_1 \to C_1 + C_2 : \quad \begin{vmatrix} a_{11} + a_{12} & a_{12} \\ a_{21} + a_{22} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Add row 2 into row 1:

$$R_1 \to R_1 + R_2 : \quad \begin{vmatrix} a_{11} + a_{21} & a_{12} + a_{22} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Same determinant both times — the extra term always cancels.This works for any scalar k i.e. kRi added to Rj OR kCi added to Cj
Anton & Rorres, p.90 (8th Ed.) / p.88 (7th Ed.); holds for columns too.

6. Scaling One Row or Column

If B results from multiplying a single row or column of A by a scalar k, then:

$$\det(B) = k\det(A)$$

Check it directly — factor k out of the row it multiplies:

$$\begin{vmatrix} ka_{11} & a_{12} \\ ka_{21} & a_{22} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = k\det(A)$$

Watch the difference: scaling the whole n×n matrix by k scales every row, so det(kA) ≠ k·det(A) in general —

$$\det(kA) = k^n \det(A) \quad \text{for an } n \times n \text{ matrix}$$

7. Proportional Rows or Columns Also Give Zero

If a square matrix has two proportional rows or two proportional columns, then det(A) = 0.

Column 2 is exactly twice column 3:

$$\det\begin{bmatrix} -2 & 8 & 4 \\ 3 & 2 & 1 \\ 1 & 10 & 5 \end{bmatrix} = 0 \quad (C_2 = 2C_3)$$

Once you spot a proportional pair of rows or columns, you already know the answer is zero — no need to expand anything.

Anton & Rorres, p.92 (8th Ed.) / p.89 (7th Ed.)

8. A Row or Column of Zeros Forces det(A) = 0

1 If a square matrix A has a row of zeros or a column of zeros, then det(A) = 0.

Check for the 2×2 case:

$$\begin{vmatrix} a & 0 \\ b & 0 \end{vmatrix} = \begin{vmatrix} a & b \\ 0 & 0 \end{vmatrix} = 0$$

A zero row or column always kills every term in the expansion — this holds at any size, not just 2×2.

2 If A is any square matrix with a row or column of zeros, A is singular (not invertible).

Anton & Rorres, p.89 (8th Ed.) / p.86 (7th Ed.) — this is the same conclusion as the identical-row/column case: some structural redundancy always forces det(A) = 0.

QUICK REFERENCE

Eight Properties at a Glance

1. **$\det(AB) = \det(A)\det(B)$**
   for same-size square matrices

2. **$\det(I) = 1$**
   identity matrix, any order

3. **$\det(A^{-1}) = 1 / \det(A)$**
   when A is invertible

4. **$\det(A) = \det(A^{T})$**
   transpose changes nothing

5. **Identical rows/columns ⇒ det = 0**
   structural redundancy

6. **Adding rows/cols together ⇒ no change**
   $R_i \to R_i + R_j$ or $C_i \to C_i + C_j$

7. **Scale one row/col by k ⇒ det scales by k**
   not the whole matrix

8. **Zero row/col, or proportional rows/cols ⇒ det = 0**
   A is singular

3.2 TWO PROPERTIES THAT NEED CARE

Beyond the Basics

Determinants are well-behaved under multiplication, but addition is a trap — and swapping rows has a sign consequence that's easy to forget. Two properties worth slowing down for.

Determinants Aren't Additive

In general, det(A) + det(B) ≠ det(A + B) — even though determinant is well-behaved under multiplication, it is not linear under addition.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \quad A + B = \begin{bmatrix} 4 & 3 \\ 3 & 8 \end{bmatrix}$$

$$\det(A) = 1, \quad \det(B) = 8, \quad \det(A + B) = 23$$

$$\det(A) + \det(B) = 9 \neq 23 = \det(A + B)$$

Anton & Rorres, p.93 (7th Ed.) / p.96 (8th Ed.) — a useful counterexample to keep in your back pocket.

9. The One Case Where It Does Add

If A and B differ in only one row (say row i), the determinant is additive in that row alone:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{vmatrix}$$

The Addition Rule

If A, B, and C are n×n matrices that differ only in row i, and row i of C equals (row i of A) + (row i of B), then:

$$\det(C) = \det(A) + \det(B)$$

The same result holds for columns.

10. Swapping Rows or Columns Flips the Sign

If B results from interchanging any two rows (or any two columns) of A, then:

$$\det(B) = -\det(A)$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix}$$

$$\det(A) = a_{11}a_{22} - a_{12}a_{21} = -(a_{12}a_{21} - a_{11}a_{22}) = -\det(B)$$

This extends directly to larger matrices — interchanging any two rows of a 3×3 (or n×n) determinant also negates it:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 7 \quad \Rightarrow \quad \det \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix} = -7$$

TRY IT YOURSELF

Practice 1 — Zero by Inspection

Without expanding anything, decide which determinants must equal zero — and say which property tells you so.

(a)

$$\begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix}$$

(b)

$$\begin{vmatrix} 3 & -1 & 2 \\ 0 & 0 & 0 \\ 5 & 7 & 1 \end{vmatrix}$$

(c)

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix}$$

Hints: check for proportional rows, a row of zeros, and identical rows — in that order, one property applies to each matrix above.

TRY IT YOURSELF

Practice 2 — Reasoning From One Fact

Suppose A is a 3×3 matrix and you're told only this:

$$\det(A) = 5 \quad (A \text{ is } 3 \times 3)$$

Using only today's properties (no expansion needed), find:

- det(2A) — careful: this scales all three rows, not just one.
- det(A^T)
- det(A^{-1})
- det(B), where B is A with two rows swapped

Every answer here is a direct one-line application of a property from today — no computation of A itself required.

TRY IT YOURSELF

Practice 3 — Practicing the determinant properties

Q. Evaluate the following determinant. Make sure to mention all the properties you apply at each step.

$$A = \begin{bmatrix} b+c & c+a & b+a \\ a & b & c \\ 1 & 1 & 1 \end{bmatrix}$$

ANSWER: $\det(A) = 0$

RECAP

Key Takeaways

Multiplicative

$\det(AB) = \det(A)\det(B)$, $\det(I) = 1$, $\det(A^{-1}) = 1/\det(A)$.

Transpose-invariant

$\det(A) = \det(A^T)$ — rows and columns are interchangeable for these purposes.

Redundancy ⇒ zero

identical or proportional rows/columns, or a zero row/column, force $\det(A) = 0$.

Row/column addition

adding one row or column into another leaves det unchanged.

Scaling

one row/column scaled by k scales det by k — but scaling the whole matrix scales det by k^n.

Swaps and sums

interchanging two rows/columns flips the sign; $\det(A)+\det(B) = \det(A+B)$ only along a single shared row.

NEXT TIME

Looking Ahead

Homework 4

See CANVAS and/or Notebook

Watch out!

If Practice 2 tempted you to actually compute A, that's worth revisiting — the whole point of today is answering without it.

Next Week

Evaluating & Using Any n×n Determinant

- Cofactor (Laplace) expansion: evaluate the determinant of any n×n matrix by expanding along a row or column, not just 2×2 or 3×3.
- The adjoint of A: build adj(A) from cofactors, then invert any invertible matrix with $A^{-1} = \text{adj}(A) / \det(A)$.
- Cramer's Rule: solve Ax = b directly as a ratio of determinants, $x_i = \det(A_i) / \det(A)$ — and prove why it works.

Questions?

Properties of Determinants · Lecture 5 complete

NEXT: EVALUATING ANY n×n DETERMINANT